Linux Shells - Interview Notes

What is a Shell?
A shell is a command-line interpreter that acts as an interface between the user and the Linux kernel. It accepts commands, interprets them, and executes programs.

Types of Linux Shells
1. sh (Bourne Shell): Original Unix shell.
2. bash (Bourne Again Shell): Default on most Linux systems; scripting, history, aliases, completion.
3. ksh (Korn Shell): Enterprise Unix shell combining sh and csh features.
4. csh (C Shell): C-like syntax.
5. tcsh: Improved csh with editing and completion.
6. zsh: Advanced customization, plugins, excellent completion.
7. fish: Friendly interactive shell with syntax highlighting.

Common Commands
echo $SHELL - show current shell
cat /etc/shells - list installed shells
chsh -s /bin/bash - change default shell

sh vs bash
sh: basic, portable, fewer features.
bash: advanced scripting, arrays, history, aliases, auto-completion.

Frequently Asked Interview Questions
- What is a shell?
- Difference between kernel and shell?
- Types of Linux shells?
- Which is the default shell?
- How do you check the current shell?
- How do you change the default shell?
- Difference between sh and bash?

30-second Interview Answer
A shell is a command-line interpreter that acts as an interface between the user and the Linux kernel. Common shells are sh, bash, ksh, csh, tcsh, zsh, and fish. Bash is the most widely used default shell because it supports scripting, command history, aliases, and auto-completion.